

# ANKIT KUMAR

## Senior Product Design Engineer — Wiring Harness & Electrical Systems

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### PROFESSIONAL SUMMARY

Wiring harness and electrical systems design engineer with 10+ years of experience across OEMs including Mercedes-Benz, Mahindra & Mahindra, Fiat Chrysler (Stellantis), and the Renault-Nissan-Mitsubishi Alliance. Specializes in schematic development (Capital Logic, Capital XC, PREEvision, OrCAD), 2D/3D harness routing, high-voltage architecture, and VDA450/DVP/SOR compliance from concept through SOP. Proven track record leading design teams, driving supplier change management, and optimizing harness cost, weight, and manufacturability.

### CORE SKILLS & TOOLS

**Design & CAD Tools:** Capital Logic, Capital XC, PREEvision, CATIA V5, AutoCAD, UG NX, OrCAD, Vismockup

**Domain Expertise:** Wiring Harness Design, EDS Planning, High Voltage Architecture, Power Distribution, Schematic Development, 2D/3D Harness Routing, Wire Size & Fuse Rating Calculations

**Standards & Process:** VDA450, DVP, SOR, DFMEA, DFA/DFM, Vehicle Integration, Change Management

**Leadership & Continuous Improvement:** Cross-functional team leadership, Cost optimization, Automation (Macros), Value Stream Mapping, DPM reduction, LAMA

### PROFESSIONAL EXPERIENCE

#### Senior Product Design Engineer | Mercedes-Benz Research & Development India

Jul 2022 – Present

- Lead schematic preparation using PREEvision (Vector) for MMA & MBEA-M platform, guiding a team of 5 engineers through wiring harness development from RFQ to SOP.
- Drive VDA450 requirement implementation, coordinating directly with the German counterpart (BTV) to align on schematic maturity for SOP readiness.
- Own supplier-facing change management for wiring harness, reducing schematic rework cycles by 50% time saved.
- Optimize harness length and routing through splice positioning, power distribution, and ground point placement, cutting material cost/weight by 1.2 Kg.
- Serve as component knowledge owner for DVP and SOR on instrument cluster and switch systems.
- Mentor peers on new EDS technologies and best practices, contributing to team-wide capability building.

#### Senior Design Engineer | Mahindra & Mahindra Ltd.

Jul 2021 – Jun 2022

- Owned wiring harness design and development from concept through SOP for off-road vehicle and agricultural machinery programs.
- Prepared wiring harness schematics in Capital Logic and 2D drawings in Capital XC.
- Conducted DFA/DFM analysis of CAD layouts to meet integration requirements, and led SOR preparation for switches and instrument cluster.
- Led fire and safety issue audits, resolving field and build issues during pre-production.
- Validated design and packaging using Vismockup and UG NX.

#### Senior Designer | Fiat Chrysler Automobiles (via Satyam Venture Engineering Services, a Tech Mahindra subsidiary) Dec 2018 – Jul 2021

- Integrated automotive electrical systems and troubleshoot wire harness design issues, performing risk analysis and countermeasure planning ahead of Step 5 milestones.
- Studied vehicle architecture and harness routing requirements in cross-functional collaboration with other engineering teams.
- Selected wiring harness components (coverings, connectors) based on vehicle function and routing layout.
- Converted 3D models to 2D layout drawings and developed circuit schematics using load-calculation analysis.
- Project: Belt-Starting Generator development (2.0L Jeep, 3.6L RAM 1500) — partnered with suppliers and manufacturing teams to meet design robustness and market standards; delivered daily/weekly test-case updates to Chief Engineering; participated in supplier assessments, failure analysis, and design review meetings.

#### Engineer, Electrical Dept. (Wiring Harness Team) | Renault-Nissan-Mitsubishi Alliance

Aug 2016 – Nov 2018

- Designed and developed wiring harnesses and automotive electrical distribution systems for global Nissan Motors (Japan) programs.
- Produced detailed electrical schematics across vehicle sections; performed wire size calculations and fuse rating selection.
- Designed electronic wiring and schematic diagrams for full vehicle programs using Macro/MS Excel and Nissan's G2B designer server.

#### **Engineer | ITI Ltd. (Government of India Undertaking)**

*Aug 2015 – Jul 2016*

- Created wiring diagrams from schematic diagrams and designed electronic wiring diagrams using OrCAD Capture.
- Debugged faults on printed circuit boards.

#### **EDUCATION**

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**B.Tech, Electronics & Communication Engineering** — H.I.T. (U.P.T.U), 2008–2012, 72.84%

#### **AWARDS & CERTIFICATIONS**

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- Best Employee of the Year (Department Level) — 2024-25
- Award for Quality Improvement via DVP Macros — 2017
- Certification in Electrical Wire Routing & Harness Flattening (CATIA V5) — EDS Technologies, 2017

#### **LANGUAGES**

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English, Hindi